SILTHAUL · FEASIBILITY

SiltHaul feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First feasibility assessment, written for the kit 1.9.1 upgrade

SiltHaul feasibility assessment

SiltHaul is feasible on paper as option A (the existing design) and as option B, and the case for either rests on three things only a trial can settle: the output ratio, the shear pin release and a local fabricator's welds. Option C is not feasible with village tools. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Option A is fully calculated: 63 N each at the cranks, rope factor 5.4 and chain 4.9 (just under the 5 asked, a fix is proposed) at the 3 kN overload limit, 0.45 of recommended anchor loads, tipping margin at 10 degrees; modelled at D3 with 245 parts and no interference	High	Output ratio 0.57 against the 0.5 target rests on assumed mud and loading rates; floor friction and mud strength are estimates	SLH-STM-001, SLH-CAL-001, `cad/fidelity.json`
Materials	Stock 40 x 40 x 2 SHS, 219.1 x 3 tube, sheet and bar; polyester rope; bought UCP bearings, 08B-1 chain, rope sheaves and shackles	High	Coating life in wet, salty or contaminated mud not assessed (S15); rope bend ratio 6 at the ramp roller (S14)	SLH-STM-001, BOM
Manufacturing and assembly	A village workshop with a small welder makes the two side frames, drum, boxes and tail plate; the kit bolts together with twelve frame bolts; the build plan has making sketches and joint pictures	Medium	245 parts and 161 fasteners; weld quality and fit-up of the drum and frame are unproven; setup is at the 20 min limit	SLH-BLD-001, SLH-DWG-101 to 109
Power and controls	Hand power only: 48 W each at the crank against 75 W sustained; no electrics	High	A crank crew tires over a shift; rest allowance of 25 % assumed	SLH-CAL-001 D3, SLH-STM-001 S2
Supply of parts	Every bought part is a catalogue item from a bearing, rigging, fastener or hardware supplier; indicative prices in the BOM	Medium	Prices are indicative and not region-specific; rated shackles and a 2 t sling are needed (SLH-REG-001 G4, G5)	BOM, SLH-MKT-001
Safety and certification	No certification route identified; rope, anchor and chain are sized on a 3 kN limit with proof-load tests planned	Medium	Rope under tension near people; contaminated mud; machinery rules of the country of use to assess	SLH-REG-001, SLH-RSK-001
Option B (crew-hauled box)	Half the mass and under half the cost; no frame; but 189 N each repeated, no overload limit	Medium	Footing on wet mud (foot boards fix it on paper); ISO 11228-2 limits unchecked	SLH-OPT-001, SLH-STM-001 S18 to S20
Option C (hand slurry pump)	Flow and power work on paper at 40 mm stroke; the made pump is not buildable with village tools and needs 18 L/min of flush water	Low	HDPE manifold block, diaphragms, flap valves and debris blocking; 2.8 times the wet waste	SLH-OPT-001, SLH-STM-001 S23 to S26

Ratings: High, Medium, Low.

What would change the verdict

A timed trial on real wet silt (PR-001), a shear pin release test (PR-011) and a first build by a local fabricator would each move a Medium to High or expose a Low. All three are TRL 4 work and are not started here.